AI Integration Roadmap

AstroIllustrator | Consolidated specification v2.1 | 22 October 2026

1. Principle and priority

AI interprets intent; deterministic code computes astronomy and geometry; the renderer draws the illustration. Model-generated angles, coordinates, star maps and constellation lines are never authoritative. Ship accurate manual rendering before AI. Highest-value AI feature is natural-language configuration for both SkyChart and ScientificDiagram.

2. Stage 0–2: no AI

Stage 0: .NET/Python pipeline and source licensing. Stage 1: Orion and all core constellation chart infrastructure. Stage 2: Summer Triangle, observer/time horizon and scientific geometry templates including Polaris/zenith. These are prerequisites: prompt interpretation should call existing validated services, not create a second scientific engine.

3. Stage 3: LLM prompt to typed illustration

Example prompt: “Show Polaris above the northern horizon from Prishtina, label its altitude and zenith in Albanian.” Model returns illustrationType=ScientificDiagram, diagramType=StarAltitude, observer location, language=sq and requested annotations. Backend geocodes/resolves location using approved data, asks for UTC date/time when exact observed position is needed, validates allowed template/parameters and invokes Python. The computed result contains altitude, azimuth and explanatory notes. Use Microsoft.Extensions.AI or a compatible structured-output model SDK; schema-bound output, low temperature if supported, strict validation, timeout, cost budget, logging and manual fallback. No Semantic Kernel required for one prompt interpreter.

4. Stage 4: image generation for scenery

Generate a landscape background with blank sky and no stars, lines, planets or text. Store source prompt, model, rights notes and hash. Compose the fixed verified scientific overlay above it. A scenic image is not evidence of the real horizon profile or exact sky appearance. Label illustrative composites and avoid suggesting physically exact foreground alignment unless validated.

5. Stage 5: RAG for educational text (optional)

Use vetted astronomy references with licenses and source identifiers. Retrieve relevant passages, draft descriptions, provide citations and require editorial verification. A small curated corpus may work with ordinary search; add embeddings/vector search only if retrieval demands it.

6. Stage 6: tool calling assistant (optional)

Allow only safe tools: ListConstellations, ListDiagramTemplates, PreviewRender, GetJobStatus and ExportRender. Confirm expensive batch actions; enforce server permissions and rate limits. Multi-agent orchestration is unnecessary.

7. Test prompts and success criteria

Prompt: “Summer Triangle with Vega, Deneb and Altair” -> Asterism request with stable star IDs, not hallucinated points. Prompt: “45 degree altitude and 180 degree azimuth” -> deterministic angle diagram. Prompt: “Polaris exact altitude tonight” -> requires observer location and a concrete date/time. Prompt: “Polaris roughly from Prishtina” -> can use clearly labeled approximate educational mode. Pass when 20 representative prompts yield valid schemas, unknown concepts fail safely, no model-provided numeric values bypass computation, AI outage does not affect manual generation and scientific overlay hashes remain stable.

8. What to defer

ML.NET training, computer vision QA, autonomous multi-agent workflows, vector databases, custom fine-tuning and cloud infrastructure. They add cost without improving the initial atlas figures. Document control: This version supersedes earlier 1.0 and 1.1 scope descriptions. Keep requests, manifests, diagrams and AI integrations aligned with this

specification.

v2.1 additions: global observer support, time-aware conjunctions and AstroIllustrator naming

AI prompts for astronomical events

Support prompts such as: "Show Venus and Jupiter over the western horizon 30 minutes after sunset from Rome on 15 November 2026." AI maps bodies, observer, local date, sunset offset and west-facing field of view into a typed EventRequest. It must not assert a conjunction without checking calculated separation.

Ambiguity and factual guardrails

If user says "next conjunction" without a date, call deterministic event search over a bounded configurable interval; do not hallucinate dates. If date missing for a fixed-date view, request it. If selected bodies are below horizon, explain and offer another time/date or an explicitly labeled conceptual diagram. Sunset-relative requests require observer coordinates.

AI tools and outputs

Allowlisted tools: ResolveObserver, ResolveTimeReference, SearchEventCandidates, ComputeEphemeris, ValidateVisibility, RenderIllustration. Model outputs JSON request only; scientific services return authoritative numbers. Log prompt interpretation separately from computed results and make manual correction available.

AI delivery order

Implement global observer and event calculations before AI prompts. First AI use case is typed request extraction for an existing deterministic renderer. Artistic twilight backdrops remain separate from scientifically verified planets and constellation layers.

Scientific note: illustrative dates and prompts are examples, not verified event predictions. All published charts require computed ephemerides and editorial review.